AN INDIAN RESET

India fast-tracks its high-speed railway project by shedding its dependence on Japan and undertaking its own civil works, indigenous trainsets and European signalling

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TRAINSETS

India's B28 trainsets will launch the country's first bullet train in 2027. How they compare with Japan's next-generation E10 Shinkansen

	India (B28)	Japan (E10)
COMMERCIAL SPEED (in kmph)	250	320
MAXIMUM SPEED (in kmph)	280	360
CAPACITY*	580	700 to 750
PER TRAINSET COST*	₹250 cr.	Not Disclosed#

*Estimated
E10 will be rolled out in 2030; the current flagship E5 trainset is estimated to cost ₹1,000 cr.

Source: JR East, BEML/ICF tender documents, NHSRCL tenders

NEARLY A DECADE AFTER PRIME MINISTER NARENDRA MODI and his then Japanese counterpart Shinzo Abe presided over the groundbreaking ceremony for India's first bullet train project, the Mumbai-Ahmedabad high-speed rail corridor is finally entering the final leg of its journey.

Already three years behind its original deadline of completion by 2023, the 508-km project has gathered pace over the past year. Land acquisition is complete, statutory clearances have been secured and utility shifting finished. Gujarat is well ahead, with civil works substantially complete across much of the alignment, track-bed construction under way and overhead electrification progressing rapidly. Foundation work has been completed at eight of the 12 stations, while work continues at the remaining four stations in Maharashtra. River bridges, depots and the underground Bandra-Kurla Complex station in Mumbai are taking shape and tunnelling has begun on the 21-km underground stretch, which includes India's first undersea rail tunnel—a 7-km section beneath Thane Creek. "The first section will open in August next year," says Union railways minister Ashwini Vaishnaw. "We are now constructing about 15-16 km every month. The progress is very good." The first section he refers to is the 100-km Surat-Bilimora-Vapi stretch in Gujarat.

Yet as India's first bullet train approaches the tracks, it has quietly become a very different project from the one unveiled in September 2017. The financing remains overwhelmingly Japanese. Much else does not. For example, the tracks and bridges were already being built by Indian engineering companies to the Japanese Shinkansen (bullet train) blueprint. Now, even the trains that will run on the corridor's first operational stretch will be made in India. Most significantly, the signalling system—the technological heart of any high-speed rail (HSR) network—will be European rather than Japanese. The result: when services begin in 2027, the trains will run at around 250 kmph, slower than Japan's latest bullet trains that exceed 300 kmph. They will also be made of steel rather than the lightweight aluminium typically used in Shinkansen trainsets.

As for financing one of India's most ambitious infrastructure projects, Japan offered terms few countries could match. It extended a soft loan of about Rs 79,000 crore, covering 81 per cent of the initial estimated cost of Rs 98,000 crore, at an interest rate of just 0.1 per cent, repayable over 50 years, with a 15-year grace period. That meant India would not have to begin repayments for a decade and a half.